12.3 Conditional Probability

Conditional probability asks: given that event B has already happened, how likely is A? The formula is $P(A \mid B) = \dfrac{P(A \text{ and } B)}{P(B)}$ — restrict attention to the world where B is true, then ask what fraction of *that* world also has A. Two-way (contingency) tables make this concrete: read down a column (or across a row) to get the reduced sample space. A key fact worth remembering: $P(A \mid B)$ and $P(B \mid A)$ are generally **not equal** — order matters. When $P(A \mid B) = P(A)$, knowing B told you nothing new about A, and the events are independent (Lesson 12.2).

Examples

Example 1: Formula, straight from given probabilities

$P(A \text{ and } B) = 0.2$ and $P(B) = 0.5$. Find $P(A \mid B)$.

Solution

Identify the formula: $P(A \mid B) = \dfrac{P(A \text{ and } B)}{P(B)}$ — restrict attention to the world where B happens, then see what fraction of it also has A.

Substitute and compute: $P(A \mid B) = \dfrac{0.2}{0.5} = 0.4$.

Answer: 0.4.

Example 2: Two-way table

Of 100 students surveyed, the results are shown below. Find $P(\text{girl} \mid \text{sports})$.

Solution

	Sports	No Sports	Total
Girls	25	35	60
Boys	15	25	40
Total	40	60	100

Conditioning on "sports" restricts attention to the Sports column only (40 students).

"Given sports" means we only look inside the Sports column: 40 students total, of whom 25 are girls.

$$P(\text{girl} \mid \text{sports}) = \frac{25}{40} = \frac{5}{8}.$$

Answer: $\frac{5}{8}$.

Example 3: Cards

A card is drawn from a standard deck. Find $P(\text{king} \mid \text{face card})$.

Solution

Face cards are J, Q, K in all 4 suits: 12 face cards total. Of those, 4 are kings.

$$P(\text{king} \mid \text{face card}) = \frac{4}{12} = \frac{1}{3}.$$

Answer: $\frac{1}{3}$.

Example 4: Checking independence

$P(A) = 0.3$ and $P(A \mid B) = 0.3$. Are A and B independent?

Solution

Independence means knowing B doesn't change A's probability: $P(A \mid B) = P(A)$.

Here $0.3 = 0.3$, so yes — **independent**.

Answer: Independent.

Example 5: Dice

A fair die is rolled. Find $P(\text{even} \mid \text{greater than } 3)$.

Solution



Blue outline = the conditioning set ("greater than 3"); orange fill = also even.

"Given greater than 3" restricts the sample space to $\{4, 5, 6\}$ — only 3 outcomes now. Of those, $\{4, 6\}$ are even.

$P(\text{even} \mid > 3) = \frac{2}{3}$.

Answer: $\frac{2}{3}$.

Example 6: A conditioning event that forces certainty

A bag has 3 red and 7 blue marbles. One is drawn. Find $P(\text{red} \mid \text{not blue})$.

Solution

"Not blue" in this bag means the marble *must* be red — there's no third color. So given "not blue," red happens every single time.

$P(\text{red} \mid \text{not blue}) = 1$.

Answer: 1.

Example 7: Order matters — the reverse conditional

Using the same table from Example 2, find $P(\text{sports} \mid \text{girl})$ and compare it to $P(\text{girl} \mid \text{sports}) = \frac{5}{8}$ from Example 2.

Solution

Now condition on "girl" — restrict to the Girls *row*: 60 girls total, of whom 25 play sports.

$P(\text{sports} \mid \text{girl}) = \frac{25}{60} = \frac{5}{12}$.

Compare: $P(\text{girl} \mid \text{sports}) = \frac{5}{8} = 0.625$, but $P(\text{sports} \mid \text{girl}) = \frac{5}{12} \approx 0.417$. **These are different** — conditioning on the column vs. the row gives different restricted sample spaces.

Answer: $\frac{5}{12}$; it is not equal to $P(\text{girl} \mid \text{sports})$.

Example 8: Solve for the reverse conditional using the formula

$P(A \text{ and } B) = 0.12$ and $P(A) = 0.3$. Find $P(B \mid A)$.

Solution

Identify: this time condition on A instead of B, so divide by $P(A)$: $P(B \mid A) = \frac{P(A \text{ and } B)}{P(A)}$.

Substitute and compute: $\frac{0.12}{0.3} = 0.4$.

Answer: 0.4.

Example 9: Larger table — word problem

A survey of 80 voters found 48 said "yes" to a proposal; of those "yes" voters, 30 were under age 30. Find $P(\text{under 30} \mid \text{yes})$.

Solution

	Yes	No	Total
Under 30	30	?	?
30 or older	18	?	?
Total	48	32	80

Only the "Yes" column (48 voters) matters for this conditional probability.

Restrict to the Yes column: 48 voters, of whom 30 are under 30.

$$P(\text{under 30} \mid \text{yes}) = \frac{30}{48} = \frac{5}{8}.$$

Answer: $\frac{5}{8}$.

Example 10: Using conditional probability to test independence

In a survey of 60 teens, 36 have phones, and 24 of the phone owners are girls. Overall, 30 of the 60 teens are girls. Determine whether "having a phone" and "being a girl" are independent in this survey.

Solution

Overall rate: $P(\text{girl}) = \frac{30}{60} = \frac{1}{2}$.

Conditional rate: $P(\text{girl} \mid \text{phone}) = \frac{24}{36} = \frac{2}{3}$.

Since $\frac{2}{3} \neq \frac{1}{2}$, knowing someone has a phone *does* change the probability they're a girl — in this survey, the two categories are **dependent**, not independent.

Answer: Dependent, since $P(\text{girl} \mid \text{phone}) = \frac{2}{3} \neq P(\text{girl}) = \frac{1}{2}$.

Example 11: A "90% accurate" test isn't what it sounds like

A disease affects 1% of a population. A test for it is 90% accurate both ways (it correctly identifies 90% of people who have the disease, and correctly clears 90% of people who don't). Given a positive test result, find $P(\text{disease} \mid \text{positive})$.

Solution

Imagine a population of $10{,}000$ for concreteness.

Have the disease: 1% of $10{,}000 = 100$ people. Of these, 90% test positive: 90 true positives (and 10 false negatives).

Don't have it: 9900 people. Of these, 10% test positive anyway (false positive): $0.10 \times 9900 = 990$ false positives.

Total positive tests: $90 + 990 = 1080$.

$$P(\text{disease} \mid \text{positive}) = \frac{90}{1080} = \frac{1}{12}.$$

Answer: $\frac{1}{12}$ — only about 8%! Even an "accurate" test gives mostly false alarms when the disease is rare, because the huge healthy population produces far more false positives than the small sick population produces true positives.

NAME: _____ Geometry (FL B.E.S.T.)

Your Turn — Show Your Work!

For each problem, solve it and verify your solution. Show all steps clearly.

Problem 1: Formula

$P(A \text{ and } B) = 0.15, P(B) = 0.6$. Find $P(A \mid B)$.

Answer: **Verification:**

Problem 2: Cards

Find $P(\text{ace} \mid \text{red card})$.

Answer: **Verification:**

Problem 3: Die

Find $P(6 \mid \text{even})$.

Answer: **Verification:**

NAME: _____ Geometry (FL B.E.S.T.)

Problem 4: Independence

$P(A) = 0.5, P(A \mid B) = 0.4$. Are they independent?

Answer: **Verification:**

Problem 5: Table

50 people; 20 take the bus; 12 of the bus riders are students. Find $P(\text{student} \mid \text{bus})$.

Answer: **Verification:**

Problem 6: Die

Find $P(\text{odd} \mid \text{less than } 4)$.

Answer: **Verification:**

NAME: _____ Geometry (FL B.E.S.T.)

Problem 7: Formula

$P(A \text{ and } B) = \frac{1}{10}, P(A) = \frac{2}{5}$. Find $P(B \mid A)$.

Answer: **Verification:**

Problem 8: Cards

Find $P(\text{heart} \mid \text{red})$.

Answer: **Verification:**

Problem 9: Independence rule

Events are independent exactly when $P(A \mid B)$ equals what?

Answer: **Verification:**

Problem 10: Marbles

Bag: 5 red, 5 blue. Find $P(\text{red} \mid \text{not green})$.

Answer: **Verification:**

Problem 11: Table

80 voters; 48 said yes; 30 of the yes voters are under 30. Find $P(\text{under } 30 \mid \text{yes})$.

Answer: **Verification:**

Problem 12: Order matters

Is $P(A \mid B)$ always the same as $P(B \mid A)$?

Answer: **Verification:**

NAME: _____ Geometry (FL B.E.S.T.)

Problem 13: Formula

$P(A \text{ and } B) = 0.12$, $P(A) = 0.3$. Find $P(B \mid A)$.

Answer: **Verification:**

Problem 14: Table application

Of 120 students, 72 take art; 45 of the art students also take music. Find $P(\text{music} \mid \text{art})$.

Answer: **Verification:**

Problem 15: Challenge

A rare condition affects 2% of a population. A test is 95% accurate both ways. Using a population of 10,000, find $P(\text{condition} \mid \text{positive test})$.

Answer: **Verification:**

NAME: _____ Geometry (FL B.E.S.T.)

✓ Answer Key

1. 0.25
2. $\frac{1}{13}$
3. $\frac{1}{3}$
4. No
5. $\frac{3}{5}$
6. $\frac{2}{3}$
7. $\frac{1}{4}$
8. $\frac{1}{2}$
9. $P(A)$
10. $\frac{1}{2}$
11. $\frac{5}{8}$
12. No, not in general
13. 0.4
14. $\frac{5}{8}$
15. $\frac{19}{68}$

Great work — keep practising!